



Cyber Attacks Explained

Database Hacking

Part—7

In the previous articles in this series, we covered the common attacks encountered in infrastructure vulnerability exploitation scenarios. This month, we are going to delve into database security attacks.

The database is the heart of any website or application and hacking into it can cause serious losses. First, let us discuss the possible database-specific attacks and then talk about the various means to protect our infrastructure from them. We will also touch upon the FOSS infrastructure protection mechanism.

Database fundamentals

We all know that databases are a structured collection of data in a digital form. The structure consists of schema definitions, tables that store the data in an organised form, views that are virtual tables, stored queries to improve performance, etc. The database engine is responsible for accepting SQL query requests, executing them and providing data back to the querying computer. For this reason, the database service listens on one or

more TCP/IP ports. The service also runs very closely with the operating system, in order to gain better control on the disk and memory resources. Modern database engines provide many features such as Web services, data encryption, etc. Figure 1 shows a typical database hosting architecture for a website.

A database server may or may not use the authentication services and access controls provided by the operating system; it often uses the controls provided by the engine itself. Each user who is supposed to log in to the database to access information is bound by basic security features such as rules and roles. Database security is designed around the data's integrity as well as the database's performance. It is important to know that security that is too tight can jeopardise database performance, which may not be acceptable in a business scenario.